

5.5 The Substitution Rule

In this section, we will review a technique for integration.

Example 1

Find an antiderivative for the following functions.

1. $f(x) = (x + 1)^2$

2. $f(x) = (x + 1)^{100}$

The Substitution Rule in Indefinite Integrals

If $u = g(x)$ is a _____ function whose range is an interval I , and f is _____ on I , then

$$\int f(g(x))g'(x) dx =$$

Note: The Substitution Rule is “the inverse” of the Chain Rule.

When do we need the substitution method?

Substitution Method:

To integrate $\int f(g(x)) g'(x) dx$:

1. Substitute $u = g(x)$ and $du = g'(x) dx$ to obtain
2. Integrate with respect to ____.
3. Replace u with _____ in the result.

Example 2

Find $\int \frac{2x + 1}{x^2 + x + 3} dx$

The Substitution Rule in Definite Integrals

If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))g'(x) dx =$$

Substitution Method:

To integrate $\int_a^b f(g(x)) g'(x) dx$:

1. Substitute $u = g(x)$, $du = (du/dx)dx = g'(x) dx$, $u(a)$ and $u(b)$ to obtain
2. Integrate with respect to ____.
3. Evaluate the integral!

Example 2

Evaluate $\int_1^e \frac{(\ln(x))^2}{2x} dx$

Extra Example

What if the substitution is not immediately obvious?

Try the following examples on your own before watching the corresponding videos.

1. $\int (x + 1)\sqrt[3]{x - 4} \, dx$

2. $\int_0^{\pi/16} \tan(4x) \, dx$